WORKSHEET 1.10

Guaranteed Values: The Intermediate Value Theorem

Unit 1: Limits and Continuity

Key ideas

- **IVT.** If f is continuous on $[a, b]$ and N is between $f(a)$ and $f(b)$, then $f(c) = N$ for at least one c in (a, b).
- A justification has three parts: continuity, the inequality with numbers, and the theorem named.
- The theorem guarantees *at least one* value; it does not locate it or count them.
- To show $f(c) = g(c)$ for some c, apply the theorem to $h = f - g$.

Worked examples

Example 1 · A root in an interval

Show that $x^3 - 4x + 1 = 0$ has a solution in $(0, 1)$.

Solution. Let $f(x) = x^3 - 4x + 1$, a polynomial, so f is continuous on $[0, 1]$. $f(0) = 1$ and $f(1) = -2$, so $f(1) < 0 < f(0)$.

By the Intermediate Value Theorem there is a c in $(0, 1)$ with $f(c) = 0$.

Example 2 · Counting from a table

A function f is continuous on $[0, 9]$.

x	0	2	4	7	9
$f(x)$	5	-1	2	6	-3

What is the least number of zeros of f on $[0, 9]$?

Solution. The sign changes on $[0, 2]$, $[2, 4]$ and $[7, 9]$, but not on $[4, 7]$.

At least 3.

Example 3 · Two functions

f and g are continuous, with $f(2) = 1$, $g(2) = 6$, $f(5) = 9$, $g(5) = 4$. Show that $f(c) = g(c)$ for some c in $(2, 5)$.

Solution. Let $h = f - g$, continuous on $[2, 5]$. $h(2) = -5 < 0$ and $h(5) = 5 > 0$.

By the Intermediate Value Theorem $h(c) = 0$ for some c in $(2, 5)$, so $f(c) = g(c)$.

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Practice

CORE

1. State the hypothesis of the Intermediate Value Theorem.

2. f is continuous on $[0, 3]$ with $f(0) = -2$ and $f(3) = 7$. Which of the values 0, 5, 8 must f take on $(0, 3)$?

3. Show that $x^5 + x - 1 = 0$ has a solution in $(0, 1)$.

4. Explain why the Intermediate Value Theorem cannot be applied to $f(x) = \dfrac{1}{x-1}$ on $[0, 2]$.

EXAM LEVEL

5. Show that the equation $e^x = 4 - x^2$ has a solution in $(1, 2)$.

6. A differentiable function P has $P(1) = 12$ and $P(4) = -3$. Must there be a t in $(1, 4)$ with $P(t) = 5$? Justify.

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7. f is continuous on $[1, 6]$ with $f(1) = 4$ and $f(6) = 4$. Must f have a zero in $(1, 6)$? Explain.

8. For which values of k does the theorem guarantee that $f(x) = x^2 + x + k$ has a zero in $(0, 1)$?

Question 9 · Free-response practice

The temperature in a greenhouse, $T(t)$ °C, is a continuous function of time t hours.

t	0	5	10	20
$T(t)$	18	25	21	30

(a) Must there be a time t, $0 < t < 5$, at which $T(t) = 23$? Justify your answer.

(b) What is the least number of times in $[0, 20]$ at which $T(t) = 23$? Show your reasoning.

(c) A student claims the table shows the temperature never fell below 18 °C. Is the claim justified?

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Worked solutions

1. f is continuous on the closed interval $[a, b]$.
2. 0 and 5 (both lie between -2 and 7). Not necessarily 8.
3. $f(x) = x^5 + x - 1$ is continuous on $[0, 1]$; $f(0) = -1 < 0 < 1 = f(1)$. By the IVT there is a c in $(0, 1)$ with $f(c) = 0$.
4. f is not continuous on $[0, 2]$: it is undefined at $x = 1$.
5. Let $h(x) = e^x + x^2 - 4$, continuous on $[1, 2]$. $h(1) = e - 3 < 0$ and $h(2) = e^2 > 0$. By the IVT $h(c) = 0$ for some c in $(1, 2)$.
6. Yes. P is differentiable, hence continuous, on $[1, 4]$, and $P(4) = -3 < 5 < 12 = P(1)$. By the IVT such a t exists.
7. Not necessarily. 0 is not between $f(1)$ and $f(6)$, so the theorem guarantees nothing.
8. $f(0) = k$ and $f(1) = 2 + k$. Zero lies between them when $k < 0 < 2 + k$: $-2 < k < 0$.
9. (a) Yes. T is continuous on $[0, 5]$ and $T(0) = 18 < 23 < 25 = T(5)$. By the Intermediate Value Theorem there is a time t in $(0, 5)$ with $T(t) = 23$.
   (b) 23 lies between the endpoint values on $[0, 5]$ (18 and 25), on $[5, 10]$ (25 and 21) and on $[10, 20]$ (21 and 30). At least 3 times.
   (c) No. The table gives the temperature at four times only; between them T may have dipped below 18 and returned.

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